Chemical industry

SOME CHEMICAL REACTIONS ARE PERFORMED ON A HUGE SCALE TO PRODUCE VALUABLE SUBSTANCES.

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Many of the raw materials that humans need exist in nature. They are refined from ores or separated from mixtures such as seawater. Some compounds, however, are made in factories using chemical reactions.

The Haber process

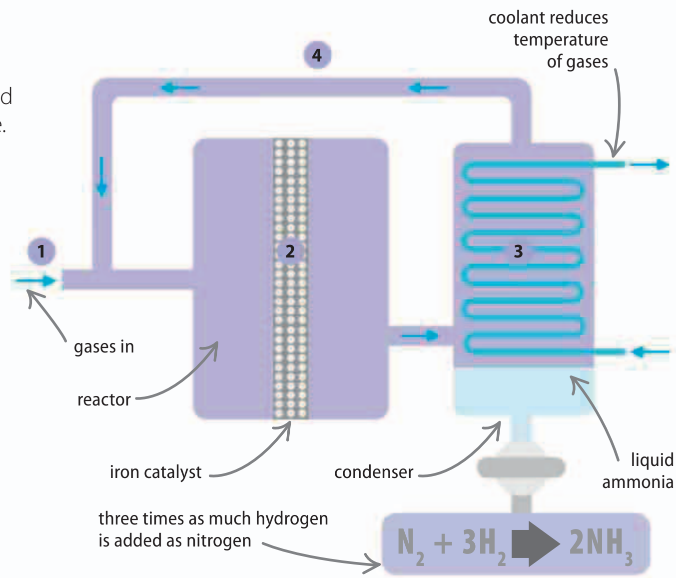
Also known by its full name, the Haber-Bosch, this process turns nitrogen and hydrogen gas into ammonia (NH_3). Ammonia is used to make crop fertilizers and explosives, such as TNT and dynamite. Nitrogen is the most common gas on Earth—it makes up 78 percent of the air—but it is very unreactive. The Haber process uses a catalyst (see page 138) to make the reaction occur.

1. Gases mixed
A mixture of hydrogen (H_2) and nitrogen (N_2) gases is pumped into the reactor. Three times as much hydrogen is added as nitrogen to create the correct ratio for ammonia (3:1).

2. In the reactor The gases are passed over an iron catalyst, which brings them together so they react to form ammonia. The reaction takes place at 450°C (842°F) and at 200 times the atmospheric pressure.

3. Product separated
The ammonia gas leaving the reactor moves into the condenser, where it is cooled so that it turns into liquid ammonia that can then be tapped off.

4. Reactants recycled Not all of the reactants turn into ammonia. The unused nitrogen and hydrogen gases rise out of the condenser and are recycled back into the reactor.



Nitric acid production

One of the chemicals that is made from ammonia is nitric acid (HNO_3). This acid reacts with bases to form nitrate salts, which plants need to make proteins. Nitric acid is mainly used to make fertilizers, but it is also used as a rocket fuel and is one of the few solvents that can dissolve gold.

1. Converter
In the converter, ammonia (NH_3) and oxygen (O_2) react at 800°C (1,472°F) using a platinum catalyst to make nitric oxide (NO) and water.

2. Oxidation chamber
The gases from the converter are cooled to 100°C (212°F). More oxygen is added, some of which will react with the nitric oxide to make nitrogen dioxide (NO_2).

3. Absorption tower
Water trickles down through the quartz crystals while the gases rise. The nitrogen dioxide and left-over oxygen react with the water to form nitric acid.